



KrishiShetra (कृषि क्षेत्र)

"Know the Price. Choose the Market. Sell Better." — Digital Ecosystem for Indian Farmers, FPOs & Transporters

Stack: Node.js • Express • MongoDB • Leaflet.js • OpenStreetMap • i18n (EN/HI/MR)

Target Ecosystem: 140M+ Indian Farmers • 10k+ FPOs • Institutional B2B Buyers

⚡ 1. The 30-Second Elevator Pitch

*"In India, 86% of farmers are smallholders who lose 25–40% of their potential income to opaque middlemen commissions, asymmetric mandi pricing, and post-harvest distress selling. **KrishiShetra** is an integrated full-stack AgriTech ecosystem that connects farmers, FPOs, verified institutional buyers, and rural transporters. Through our **real-time APMC Mandi Arbitrage Engine**, **Multi-Turn Counter Negotiation Room**, **AI 'Sell Now vs Store & Hold' Decision Engine**, and **e-NWR Pledge Financing integration**, we turn smallholder farmers into informed, profitable agri-entrepreneurs."*

🏗️ 2. High-Level Architecture & Core Value Proposition

👤 1. Farmers & FPOs

Farm-gate lot creation, 45+ mandi GPS arbitrage, AI hold/sell recommendation, multi-turn counter-offer chat.

🏢 2. Institutional Buyers

Direct verified crop procurement, custom quality specs, instant deal conversion, escrow-backed fulfillment.

🚚 3. Transporters & Storage

Return-load dispatch matching, accredited MSWC/CWC warehouse booking, and instant pledge loan unlock.

[Farmer / FPO Lot] → [Market Discovery] → [Multi-Turn Counter Negotiation] → [Deal Accepted] | [Delivered & Paid] ← [In Transit (GPS)] ← [Ready for Pickup] ← [6-Stage Order KS-ORD-XXX]

🏛️ 3. Comprehensive Judge Q&A Defense Bank

Q1: "Government already built e-NAM. Why do we need KrishiShetra? How are you different?"

MARKET & GROUND

Evaluator Intent: Testing if you understand e-NAM's actual ground friction vs theoretical premise.

Direct Answer: e-NAM is a visionary policy tool, but on the ground it suffers from 3 major operational bottlenecks that KrishiShetra specifically solves:

- 1. Pre-Transport Listing vs Mandi Yard Queue:** e-NAM still requires farmers to physically transport crops to the APMC mandi yard before bidding starts, incurring locked-in transport costs. KrishiShetra enables *farm-gate advance listing* before spending a single rupee on freight.
- 2. Dynamic Multi-Turn Counter Negotiation:** e-NAM relies on rigid single-round closed bids. KrishiShetra features an interactive counter-offer negotiation room where terms, price, and delivery timelines are refined collaboratively.
- 3. Integrated Post-Harvest Decision Intelligence:** e-NAM never advises a farmer whether to sell or store. KrishiShetra calculates biological moisture loss, warehouse rent, and seasonal price trajectories to give a quantified ROI recommendation.

Q2: "Traditional middlemen (Arhtiyas) lend cash to farmers. How will you get farmers to switch?"

AGRIFINANCE

Evaluator Intent: Testing your knowledge of the informal credit trap tying farmers to commission agents.

Direct Answer: Middlemen retain farmers not through love, but through usurious credit (24%–36% p.a.) that forces pre-harvest crop commitments. We dismantle this loop through:

- **e-NWR Warehouse Pledge Financing:** In `PledgeFinancingRequest.js`, farmers storing produce in accredited MSWC/CWC facilities immediately access **70%–75% pledge loan disbursements** from partner financial institutions at 7%–9% subsidized agri-credit rates.
- **FPO Collective Shield:** Aggregating smallholders via our `fpo-dashboard.html` allows group collateral and institutional buyer advance contracts.

Q3: "How does your AI 'Sell Now vs Store & Hold' Decision Engine calculate its recommendations?"

AI & MATH

Evaluator Intent: Verifying if your AI is an actual algorithmic model or just arbitrary placeholder code.

Direct Answer: In `server/services/decision.service.js`, our engine evaluates:

Net Gain = Projected Future Realization - (Current Realization + Total Holding Costs)

It incorporates four physiological and economic parameters:

- **Storage Tariff:** Exact facility rates (e.g., ₹38/bag/mo dry warehouse, ₹65/bag/mo cold chain).
- **Biological Weight & Moisture Shrinkage:** Crop-specific loss models (**Onion = 4.5%, Tomato = 3.8%, Potato = 2.2%, Wheat = 0.8%** over 30–60 days).
- **Handling & Freight Delta:** Loading charges (₹14/qtl) + distance to warehouse via Haversine calculation (₹14/km).
- **Threshold Gate:** A "STORE & HOLD" recommendation is only triggered if net ROI is > **+2.5%**.

Q4: "How does the Mandi Price Arbitrage Engine compute freight and net profitability?"

GEOSPATIAL

Evaluator Intent: Testing geodesic math and real transport cost reality.

Direct Answer: In js/mandi-map.js and js/mandi-compare.js:

- Calculates great-circle distance between farmer coordinates and 45+ APMC mandis using the **Haversine Formula**.
- Applies dynamic rural vehicle freight curves: $\text{Transport Cost} = \text{Distance (km)} \times \text{Qty (qt1)} \times \text{₹18-₹20/km}$.
- Computes **Net Realization** = $(\text{Modal Mandi Price} \times \text{Qty}) - \text{Freight Cost}$ and automatically marks the highest net-profit market with a 🏆 Gold Trophy.

Q5: "What is your Business Model? Who pays for KrishiShetra?"

MONETIZATION

Evaluator Intent: Ensuring you do not charge impoverished smallholder farmers.

Direct Answer: We follow a **Zero-Fee for Smallholder Farmers, Monetize B2B & Service Providers** model:

Revenue Stream	Fee Structure	Paid By
Institutional B2B Procurement Fee	1.0% – 1.5% of GMV	Corporate buyers (ITC, BigBasket, Millers) for verified direct sourcing
Freight Dispatch Take-Rate	3.0% – 5.0% per trip	Transporters for guaranteed return-load job matching
Warehouse Booking Lead Fee	₹15 – ₹25 per quintal	Accredited private & state warehouse/cold storage operators
FinTech Loan Origination Cut	0.5% – 1.0% loan value	Partner NBFCs/Banks for qualified e-NWR pledge loan leads

Q6: "How do you solve the Trust & Quality Assaying problem without seeing the crop?"

QUALITY & TRUST

Evaluator Intent: Understanding dispute mitigation and remote commodity trade risks.

Direct Answer: We enforce a 3-layer trust architecture:

- **AGMARK Grade Parameters:** Mandatory lot attributes (Grade A/B/C, moisture %, harvest date, seed variety, visual photos).
- **Accredited Warehouse Assaying:** When stored in partner facilities, government-certified lab assaying certificates are attached to the e-NWR lot.
- **6-Stage Escrow Pipeline:** Buyer funds are placed in escrow upon order confirmation and released to farmer/transporter only upon destination OTP delivery sign-off.

4. Deep-Dive Ground Case Studies

CASE 1: Nashik Onion Farmer — AI "Sell vs Store" + Pledge Loan

Dindori, Maharashtra

BASELINE DISTRESS SALE (AT HARVEST - MARCH)

- Farmer: Ramesh Patil (100 Quintals / 200 Bags Red Onion)
- Local Mandi Modal Rate: ₹1,200 / quintal
- Gross Realization: ₹1,20,000
- Middlemen Comm. & Deductions: -₹8,000
- **Net Take-Home: ₹1,12,000**

KRISHISHETRA AI RECOMMENDATION (60-DAY COLD STORE)

- Projected June Price: ₹2,100 / quintal (+75%)
- Storage Rent (Sahyadri Cold Chain @ ₹65/bag/mo): -₹26,000
- Shrinkage / Moisture Loss (4.5% over 60d): -₹9,450
- Handling & Local Transport (12 km): -₹3,500
- **Net Realization: ₹1,71,050 (+₹59,050 / +52.7% Extra Gain)**

*Unlocked ₹80,000 instant pledge loan for seasonal working capital.

LOCAL MANDI (DEWAS APMC - 8 KM)

- Produce: 80 Quintals Yellow Soybean
- Modal Rate: ₹4,400 / quintal
- Transport: ₹160 (local tractor)
- **Net Profit: ₹3,51,840**

ARBITRAGE MANDI (INDORE APMC - 38 KM)

- Indore Modal Rate: ₹4,850 / quintal (+₹450/qtl higher)
 - Freight Cost (Mini-Truck @ ₹18/km): -₹684
 - **Net Profit: ₹3,87,316 (+₹35,476 Net Day Profit)**
- *Identified via KrishiShetra GPS Mandi Comparison in 10 seconds.

🌟 5. Scope of Improvement & Future Roadmap

Phase 1: 0–3 Months

- Smartphone Camera AI Quality Assaying (CV-based grain grading).
- Vernacular Voice Bot (Whisper/Bhashini voice bidding in Hindi/Marathi).
- WhatsApp Business Bot for daily alerts.

Phase 2: 3–6 Months

- Reefer Cold Truck IoT Telematics (real-time temperature logs).
- ONDC Agri-Network interoperable buyer protocol.
- Automated UPI Auto-Split escrow payouts.

Phase 3: 6–12 Months

- Satellite Remote Sensing & NDVI Indexing for farm yield validation.
- Parametric Weather Insurance automatic smart contracts.
- Micro-warehouse franchise network.

🎯 6. Tough "Grilling" Questions Quick-Ref

Jury Trap Question	Underlying Concern	Winning Response
"What if there is no internet in rural areas?"	Offline usability and edge resilience.	PWA offline caching for recent mandi prices + SMS/IVR dial-in fallback protocol.
"What if a buyer rejects produce at delivery?"	Trade default and settlement risks.	Escrow hold contract + standard APMC assaying tolerance thresholds ($\pm 2\%$ moisture variance).
"How will you acquire remote truckers?"	Two-sided logistics liquidity.	Focusing on <i>Return-Load Matching</i> —turning empty return trips into high-margin revenue.
"Are AI forecasts accurate during weather shocks?"	ML integrity and statistical honesty.	Engine outputs confidence ratings (70%–88%) and flags weather anomalies with risk alerts.

🏆 **Presentation Golden Rule:** Lead with farmer impact (e.g. Ramesh Patil's ₹59k extra income), demonstrate the live tri-lingual UI (Hindi/Marathi), and position KrishiShetra as complete digital market & financial infrastructure.